

Gotta go the biotech way

Srikanth R

readersletters@thinkdigit.com

“Biotech” (BT) conjures an image of cutting edge research undertaken in sterile high tech labs, the products of which rarely concern the common man. But, contrarily, BT has been in the public consciousness for many reasons and also since antiquity – that belies its cutting edge image. For example, the common process of converting milk to curd is BT, or the process of making *dosa* batter sour.

In general, “technology” refers to the application of science. In the same vein, Biotech or Biotechnology is “any technological application that uses biological systems, living organisms, or derivatives thereof, to make or modify products or processes for specific use” – as defined by the UN. The “biotech” in the conversion of milk to curd involves the use of bacteria to break down milk protein into lactic acid.

What is BT?

BT is not a discrete field of study, rather a convenient association of different sciences with biology. This association is driven by the need to find greener alternatives to present processes. And it is evident that the thrust of BT has two directions: firstly to discover organisms that can do biologically what is presently being done chemically, and secondly to modify existing biological agents to improve their efficiency. While there are terms such as industrial BT, textile BT, medical BT, food BT, agricultural BT.

etc., to describe the various associations, a more meaningful division could be: red BT, denoting all BT activities that relate to medicine/healthcare; green BT, for all agriculture/food related BT activities; blue BT, includes all activities relating to aquatic applications of BT; and, white BT, referring to all activities relating to BT applications in various industries. We shall see how each of these BT subclasses has impacted us.

Applications White BT

The Industrial Revolution, while improving lives, has destroyed environments. Stringent environmental laws and greater awareness have meant that industries need to treat their wastes before disposal, a cost that effects their bottomlines. This has spurred the search for greener alternatives. This search has zeroed in on enzymes. Enzymes are chemicals that are naturally generated by living beings to break down food. Their action is similar to those of catalysts in normal chemical reactions; and being biodegradable they don't need further treatment before disposal. There are different types of enzymes, and they are classified based on their target molecule. For example lipases are enzymes that aid the breakdown of Lipids or oily substances, cellulases are enzymes that aid the breakdown of cellulose, etc. BT has exploited this natural ability in micro-organisms (MOs), and in many cases enhanced or modified this ability to serve mankind. Some places where BT is applied:

* Textile industry

Enzymes are increasingly being used in different processes instead of chemicals.

For example, hydrogen peroxide, which is used to bleach cotton, needs to be removed from the fabric before it can be dyed. There are peroxidase enzymes which can do the same job, saving a lot of water which would otherwise have to be used for rinsing. Enzymes also play a role in creating the “stone washed” look in jeans – this was originally created using pumice stones, but no longer. Cotton, being white, needs to be dyed to create textiles of any other colour, requiring a lot of chemical dyes. BT cottons that have been genetically modified to express colour makes the process of dyeing redundant.

- Textile cleaning industry
Stubborn stains can be broken down by specific enzymes; for example, stains left by oily substances can be broken down using lipases. Enzymes that work well at normal room temperatures obviate the need for high temperature and harsh chemicals for the job. In India, premium detergents like Ariel and Surf Excel include enzymes to tackle tough stains without damaging the cloth (or the environment).

- Paper industry
Converting wood to paper requires removing lignin from the wood. Fungi produce the laccase enzyme that can breakdown lignin, and is ideal for this job. Another case: in order to make used paper fit for reuse, it needs to be free from glues or other sticky materials. Esterase enzymes that attack the bonds that allow the sticky substance to cling to the pulp are used for the separation. Traditionally, both these steps require treating the pulp with chemicals at a high temperature. With enzymes, the process